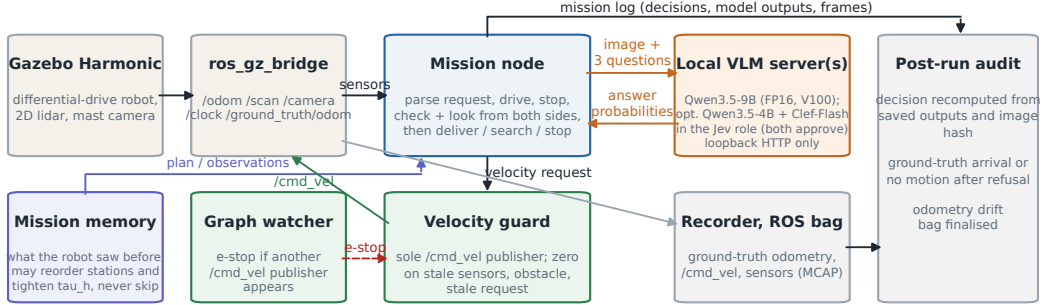
mission log (decisions, model outputs, frames)
Gazebo Harmonic
differential-drive robot, 2D lidar, mast camera
ros_gz_bridge
/odom /scan /camera /clock /ground_truth/odom
sensors
Mission node
parse request, drive, stop, check + look from both sides, then deliver / search / stop
image + 3 questions
answer probabilities
Local VLM server(s)
Qwen3.5-9B (FP16, V100); opt. Qwen3.5-4B + Clef-Flash in the Jev role (both approve) loopback HTTP only
Post-run audit
decision recomputed from saved outputs and image hash
ground-truth arrival or no motion after refusal
odometry drift bag finalised
plan / observations
/cmd_vel
velocity request
Mission memory
what the robot saw before may reorder stations and tighten tau_h, never skip
Graph watcher
e-stop if another /cmd_vel publisher appears
e-stop
Velocity guard
sole /cmd_vel publisher; zero on stale sensors, obstacle, stale request
Recorder, ROS bag
ground-truth odometry, /cmd_vel, sensors (MCAP)